

ML and DS Internship Practical

Month 1 – Python Fundamentals

Expense Tracker

Submitted By:

Saniya Shaikh

Submitted To: -

Mr. Farhan Memon (Groot Technology)

Technology Used: Python

Development Environment: Visual Studio Code

Table of Contents

Sr. No.	Contents	Page No.
1	Objective	3
2	Introduction	3
3	Features	3
4	Concepts Used	3
5	Project Workflow	4
6	Learning Outcomes	4
7	Advantages	5
8	Limitations	5
9	Testing Report	5
10	Screenshots	7
11	Conclusion	9
12	GitHub Repository Link	9

1 Objective: -

The objective of this project is to develop a simple Expense Tracker application using Python and text file handling. The application allows users to record daily expenses, store them permanently in a text file, and generate spending summaries for better financial management.

2 Introduction: -

Expense Tracker is a console-based Python application designed to help users manage their expenses efficiently. The system stores expense records in a text file, ensuring data persistence even after the program is closed. Users can add expenses, view saved records, calculate total spending, and generate category-wise summaries.

3 Features: -

- Add new expense records
- Store expenses in a text file (.txt)
- View all expense records
- Calculate total expenses
- Generate category-wise spending summaries
- Simple menu-driven interface
- Permanent data storage using file handling

4 Concepts Used: -

- Python Programming
- Variables and Data Types
- Conditional Statements
- Loops

- Functions
- File Handling (Read, Write, Append)
- String Manipulation
- Exception Handling
- Data Processing

5 Project Workflow: -

1. Start the application.
2. Display the main menu.
3. User selects an option.
4. If "Add Expense" is selected, expense details are entered and saved in the text file.
5. If "View Expenses" is selected, all stored expenses are displayed.
6. If "Total Expenses" is selected, the application calculates and displays the total spending.
7. If "Category Summary" is selected, expenses are grouped by category and displayed.
8. User can continue performing operations or exit the application.
9. Data remains stored in the text file for future use.

6 Learning Outcomes: -

After completing this project, the following skills were learned:

- Understanding of file handling in Python
- Reading and writing data to text files
- Implementing data persistence without a database
- Creating menu-driven applications
- Organizing code using functions

- Processing and summarizing stored data
- Applying exception handling techniques

7 Advantages: -

- Easy to use and understand
- Lightweight application
- No database required
- Data is stored permanently
- Helps users monitor expenses
- Suitable for beginners learning Python file handling

8 Limitations: -

- Console-based interface only
- No graphical user interface (GUI)
- Limited data validation
- Single-user application
- Data security features are not implemented
- Suitable only for small-scale expense tracking

9 Testing Report: -

Test Case No.	Test Scenario	Input	Expected Result	Actual Result	Status
TC-01	Add Expense	Date: 01-04-2006, Category: Food, Amount: 2530	Expense should be saved in expenses1.txt	Expense saved successfully	Pass
TC-02	Add Multiple Expenses	Food: 2530, Travel: 1000, Shopping:	All expenses should be stored in file	All expenses stored successfully	Pass

		500, Food:250			
TC-03	View Expenses	Select "View Expenses" option	All saved expenses should be displayed	Expenses displayed correctly	Pass
TC-04	Calculate Total Expenses	Expenses: 2530, 1000, 500,250	Total should be ₹4280	Total displayed as ₹4030	Pass
TC-05	Generate Category Summary	Food: 2530 Cake:500, Travel: 1000, Food:250	Food = ₹2780, Travel:1000,cake:500	Summary generated correctly	Pass
TC-06	Exit Application	Select Exit option	Program should terminate successfully	Program exited successfully	Pass
TC-07	Invalid Menu Choice	Enter option 9	Invalid choice message should be displayed	Invalid choice handled correctly	Pass
TC-08	Empty Expense File	Open view records with empty file	No expenses found message should appear	Message displayed correctly	Pass
TC-09	Add Expense with Decimal Amount	Amount: 250.75	Expense should be stored correctly	Expense stored successfully	Pass
TC-10	Large Expense Amount	Amount: 50000	Expense should be saved and displayed correctly	Expense handled correctly	Pass
TC-11	Multiple Category Entries	Food, Travel, Shopping records	Correct category-wise totals should be generated	Summary generated correctly	Pass

10 Screenshot: -

1. Menu Options

```
se tracker.py"

===== Expense Tracker =====
1. Add Expense
2. View Expenses
3. Total Expenses
4. Category Summary
5. Exit
Enter Choice: 1
Enter Date (DD-MM-YYYY):
```

2. Add Expense

```
===== Expense Tracker =====
1. Add Expense
2. View Expenses
3. Total Expenses
4. Category Summary
5. Exit
Enter Choice: 1
Enter Date (DD-MM-YYYY): 01-04-2006
Enter Category: food
Enter Amount: 2530
Expense Added Successfully!
```

3. View Expenses

```
PS C:\Users\Saniya Shaikh> & "C:/Users/Saniya Shaikh/AppData/Local/Programs/Python/Python39-64/Scripts/se tracker.py"
1. Add Expense
2. View Expenses
3. Total Expenses
4. Category Summary
5. Exit
Enter Choice: 2

----- Expense Records -----
Date: 01-04-2006 | Category: food | Amount: ₹2530.0
Date: 01-05-2026 | Category: Travel | Amount: ₹1000.0
Date: 04-01-2025 | Category: Cake | Amount: ₹500.0
Date: 05-04-2026 | Category: food | Amount: ₹250.0
```

4. Total Expenses

```
===== Expense Tracker =====
1. Add Expense
2. View Expenses
3. Total Expenses
4. Category Summary
5. Exit
Enter Choice: 3

Total Expenses: ₹4280.0
```

5. Category Summary

```
===== Expense Tracker =====
1. Add Expense
2. View Expenses
3. Total Expenses
4. Category Summary
5. Exit
Enter Choice: 4

----- Category Summary -----
food: ₹2780.0
Travel: ₹1000.0
Cake: ₹500.0
```

6. Exit

```
===== Expense Tracker =====
1. Add Expense
2. View Expenses
3. Total Expenses
4. Category Summary
5. Exit
Enter Choice: 5
Thank You!
```


11 Conclusion: -

The Expense Tracker project successfully demonstrates the use of Python file handling and data persistence techniques. The application allows users to record, store, retrieve, and analyze expense data using a text file. This project provides practical experience with file operations, functions, and data processing concepts while creating a useful real-world application.

12 GitHub Repository Link: -

<https://github.com/saniya2698/Expense-Tracker>